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Artificial intelligence in education: A text mining-based review of the past 56 years

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Abstract
Artificial Intelligence in Education (AIED) is a broad and multifarious area of study that spans across various academic fields. Due to the high numbers of studies in this field, it seems too challenging to analyze all of them in depth in a single study. Additionally, there is a lack of research that provides a comprehensive overview of the main trends and topics in AIED. This study, hence, aims to fill this gap by using text mining techniques to examine how artificial intelligence (AI)-related research in education has evolved over time. To this end, a total of 11,027 articles indexed by the Scopus database in the field of education between 1967 and 2023 were examined. Based on the findings, there has been a significant increase in AIED since 2014, covering 73% of the publications. Over the past three decades, AIED research has increasingly concentrated on engineering student populations and conference proceedings. Notably, AI solutions are extensively employed in education, with a strong focus on personalization, assessment, and evaluation. They also play a prominent role in research review processes, such as text mining and topic modeling for summarizing research findings. The findings contribute to the field, enhancing our understanding of the patterns of AI's integration into education and offering guidance for prospective research endeavors.

Keywords Artificial Intelligence in Education (AIED) · Text mining · Topic modeling · Latent Dirichlet allocation · AIED

1 Introduction

On the day this manuscript was written (July 24, 2023), an article in *The Guardian* attracted our attention. The article reported that Hollywood writers protested against AI, considered technology as a threat for themselves, and did not want to be replaced by ChatGPT (Beckett, 2023). Despite the heated debates over the use

Extended author information available on the last page of the article of AI in our lives, it has been with us since 1956, when a two-month study was carried out by John McCarthy and his colleagues during the summer of 1956 at Dartmouth College in Hanover, New Hampshire (McCorduck et al., 1977). Since then, there have been significant developments in computer technologies and IT globally. The use and efficiency of AI in various sectors, including education, have significantly improved. In 1955, AI meant "making a machine behave in ways that would be called intelligent if a human were so behaving" (McCarthy et al., 1955, p.1). However, recent definitions of AI encompass cognitive processing similar to the function of the human brain and its data organization that deals with complex tasks (Karsenti, 2019; Popenici & Kerr, 2017). With this function, AI technologies have been applied to transportation, healthcare, public safety, employment, entertainment, service robots, smart cities, and education (Cantú-Ortiz et al., 2020), and AI technologies are closely related to recent economic growth (Wang & Li, 2023). AI has the potential to contribute significantly to learning and instruction, so the number of studies into the use of AI tools and applications in education is likely to increase considerably (Paek & Kim, 2021; Zhai et al., 2021).

1.1 Relevance of AI in the educational context

Though the use of emerging technology has made significant contributions to education, AI seems to be more transformative in macro aspects of education, such as teaching quality and administrative support (Ullah et al., 2022). This is fairly clear, especially when we see that AI offers opportunities and challenges for education, with the potential to cause profound changes in the "structure, operation, and governance of educational institutes" (Popenici & Kerr, 2017, p. 22), or even a revolutionary change in education, such as the Fourth Industrial Revolution (Dai et al., 2020; Knox, 2020). It positively influences educational processes by making education "highly interactive, highly accessible, and highly individualized in nature" (Karan & Angadi, 2023, p.3). Moreover, the United Nations Sustainable Development Goal 4 targets improving the quality of education and Goal 1 deals with eliminating poverty (United Nations, 2023). Thus, if AI tools and applications could be used to increase the quality of education, then they may contribute significantly to good health, for education increases well-being (Gibbons & Silva, 2011), and employment (e.g., betterment of skills development through AI-based simulations). Quality education through AI tools, regardless of location, may make it easy to find decent jobs, which is believed to reduce poverty.

AI research has mainly dealt with intelligence regarding its relation to learning, reasoning, problem solving, perception, and language use (Pedro et al., 2019). These processes are also scientifically investigated by researchers in education. Thus, AI could help improve processes such as learning, reasoning, and problem solving, which puts AI in the educational context. On the other hand, education and learning do not occur in isolation; that is, delivery of educational content requires social context and organizational structure. Hence, considering the functional improvement in educational services through AI tools and applications, AI facilitates educational processes at academic (assessment, feedback, tutoring, etc.) and administrative levels (admission, counseling, library services, etc.) (Ahmad et al., 2022).

AI relies on big data, cloud computing, artificial neural networks, and machine learning, all of which have brought about machine intelligence (Zhai et al., 2021). However, the use of AI techniques, applications, and algorithms vary from sector to sector. In their systematic literature review, Aljarrah et al. (2021) showed that the most common techniques used in distance education were deep learning, machine learning, and affective computing. In their study into the adaptive systems used in education, Colchester et al. (2017) mentioned Fuzzy Logic (FL), Decision Trees, Bayesian Networks, Neural Networks, Genetic Algorithms, and Hidden Markov Models as AI techniques for adaptive systems, the focus of which changes from identifying "student characteristics to generating profiles of the students with the intention of evaluating their overall level of knowledge to be used as a basis for prescribed software pedagogy" (p.53). In another systematic review, Ouyang et al. (2022) demonstrated how the quality of learning and instruction can be enhanced by using Sequential Pattern Mining (SPM) for identification and recommendations in learners' learning sequences; Genetic Algorithms (GA), Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) for content optimization; Machine Learning (ML) (e.g., feed-forward neural networks, support vector machines and probabilistic ensemble simplified fuzzy ARTMAP) for academic predictions; and Natural Language Processing (NLP) for code detection or emotional analysis.

Using different types of educational technologies entails matching philosophical and pedagogical approaches, which changes the way learners acquire information and the way teachers deliver instruction (Hwang et al., 2020; Su et al., 2023a). However, it is not often the case that we encounter educational and learning theories in the studies on AIED; thus, there is a gap to bridge between AI tools, applications, and educational theories (Chen et al., 2020b; Hwang et al., 2020; Hwang & Tu, 2021). As Vázquez-Cano (2021, p.7) puts it, "One of the prominent emerging challenges in education consists of proposing models and propositions for the integration of AI into teaching and learning processes, based on solid didactic and pedagogical principles. Meeting this challenge appropriately and effectively may help to create more flexible, personalized, and sustainable learning environments." Still, Ouyang et al. (2022, p.2) suggest three paradigms in the use of AI in learning and instruction. Paradigm one is "AI-directed" and considers "learner-as-recipient." Here AI is employed for representing and directing learning, as the learners are relatively passive recipients. Paradigm two is "AI-supported," in which the learner is regarded as collaborator. Here AI functions as a support for students' learning as the learners work together with AI. Paradigm three is "AI-empowered," in which AI is for empowerment while the learner is in the leader position, "taking agency of their learning." Consequently, despite the technological developments and improvements in AI systems, it seems that there is much to do to make pedagogical practices compatible with philosophical approaches in AIED as the way of learning and the structure of instruction in the new world of AI are still being disputed (Carvalho et al., 2022; Peters et al., 2024; Tesar et al., 2022).

1.2 The role of AIED implementations

AI is used in education in diverse ways according to the target of its implementation. It is most commonly used for personalizing learning and supporting students with special needs (Hopcan et al., 2023). It offers advantages for online and face-to-face courses, such as the assessment of a large number of assignments, detection of learning and teaching gaps, effective measuring of student progress using digital profiles, grading of papers and evaluation of exams, opportunities to study in and outside of the classroom, smart school, and mobile remote education (Chassignol et al., 2018; Chen et al., 2020a). Unlike conventional schools, a smart school makes use of advanced technology and equipment in classrooms to give students better and more effective learning experiences (Umeri, 2021), such as Smart School Emergency Tents (SADAR) which are modular-based tents for post disaster recovery in education. "This modular tent uses polycrystalline type solar panels as its power source and a knock-down multifunctional portable table that can be used as a table and bed" (Farihah et al., 2024, p.1). Moreover, AI can be used in education for learning, instruction, and administration. For learning, it could help make early diagnosis of students' learning difficulties, "predict the career path for each student" by collecting data about their learning processes, detect their existing level of achievement, and apply intelligent adaptive intervention to students (Chen et al., 2020a, p.75272). For instruction, AI could help predict student behavior, go through course content and customize it when necessary, and plan activities for learning outside the classroom. For administration, it could facilitate and accelerate the operation of the administrative duties that normally consume too much of instructors' time and identify the needs of students (Chen et al., 2020a).

AI has been used in education in applications such as teaching robots, intelligent tutoring systems (ITS), learning analytics dashboards, adaptive learning systems, and human–computer interactions (Chen et al., 2020b; Luckin et al., 2016). Moreover, ITSs, predictive analytics, automated grading, chatbots, and virtual assistants are commonly used in higher education in the form of personalized learning tools (Al Ka'bi, 2023).

1.3 AIED outcomes

AI outcomes in education are mainly eight-fold (see Fig. 1): The first is advanced personalized learning paths in four different channels: monitoring students' input, delivering appropriate tasks, providing effective feedback, and applying interfaces for human–computer communication (Zhai et al., 2021). This means that through AI algorithms, individual student performance can be effectively monitored, taking learning styles into account. Further, personalized learning paths provide opportunities to dynamically adjust the points at which content difficulty arises.

The second is early identification of learning problems and possibility for intervention. AI systems can enable us to discover the shortcomings in students' learning and address them early in education (Chen et al., 2020a). If early intervention strategies can be implemented effectively, dropout rates may be reduced and academic

Fig. 1 AIED outcomes

success can be improved. The third is interactive learning experiences. The application of AI into education has made lessons highly interactive, with interactive tutoring being the best example (Cameron, 2019; Karsenti, 2019). The fourth is enhanced collaborative learning (Luckin et al., 2016). It is now possible with AI systems to connect students with similar interests across different locations to engage in collaborative projects. The fifth is automated curriculum development and content production. AI can contribute significantly to curriculum design by analyzing students' educational needs and existing trends, making adaptations when changes occur (e.g., curriculum design using AI back propagation method by Somasundaram et al. (2020)). The sixth is immediate (real time) feedback and support. With an artificial neural network, immediate feedback according to students' input can be provided to assist them with their homework and in grasping abstract concepts (Vattam et al., 2011). The seventh is facilitating inclusive education. With AI tools and applications, it is now easier to address the educational needs of learners with special needs

(Hopcan et al., 2023) (e.g., text-to-braille converters for visually impaired students). The eighth is teacher-related and student-related outcomes. Xia et al. (2022) carried out a systematic literature review and examined the literature in the last 10 years (2012–2021) using matrix coding and content analysis approaches. They found teacher-related outcomes of AI tools and applications in education such as working efficiency, teaching competence, and attitudes towards AIED and considered student-related outcomes such as motivation and engagement, academic performance, twenty-first century skills, and non-cognitive aspects.

1.4 Theoretical perspectives of AIED

AIED research seems to investigate AI's impact on education from a technological perspective rather than an educational one (Xu & Ouyang, 2022). Hwang et al. (2020) emphasize the importance of developing AI-based learning models or application frameworks, taking both new technologies and learning theories into account.

It is noted that researchers in the field of AIED can integrate appropriate pedagogical approaches into AI applications by basing them on existing learning theories (Hwang et al., 2020). In addition, AIED is a complex adaptive system in which the interaction between pedagogical, social, and cultural factors is critical. From the perspective of complex adaptive systems theory, AIED has influenced the social relationships in educational environments (Ouyang & Jiao, 2021).

Teacher-student relationships AIED explores the shift of teacher-student relationships from a position where the teacher is the classroom authority to the teacher's position as a guide, facilitator, and collaborator for the students who have difficulty in learning in the AI-enabled environment (Xu & Ouyang, 2022).

Student's relationship with self AI systems tend to be learner-centered, encouraging individuals to reflect on their own learning (Ouyang & Jiao, 2021). At this point, a perspective based on cognitive constructivism prevails, according to which "knowledge is said to be constructed, or actively structured, by a knower's mental processes even as particular purposes and contextual constraints guide the result" (Hruby & Roegiers, 2012, p.1). AIED emphasizes the student's self-relationship, which requires students to take initiative in the construction of knowledge, rather than being passive recipients of information. While this process is usually carried out with teacher support, AIED applications, such as the smart teaching system, provide more support for students in building these cognitive structures (Xu & Ouyang, 2022).

Student–student relationship AIED bases the relationship between students on a perspective built on social constructivism (Ouyang & Jiao, 2021; Xu & Ouyang, 2022), which argues that reality is constructed through human activity; knowledge is a human product that is socially and culturally constructed, and learning occurs in a social process (Amineh & Asl, 2015). Specifically, social constructivism in AIED "reflects a notion that learning occurs when a learner interacts with people, information, and technology in socially situated contexts" (Ouyang & Jiao, 2021, p.3).

When artificial intelligence is applied in education, the nature of the student–student relationship is affected by the way students interact with AI systems as another source of social effect. In this way, while interacting with each other, students are influenced by AI systems which now have a role in their learning as students collaborate with them. Finally, because AI (or emotional artificial intelligence) can recognize emotions by using recurrent neural networks in social settings (Assunção et al., 2022; Kambur, 2021; Loghmani et al., 2017), it can facilitate the understanding of complex social group relationships and help teachers manage their classrooms more easily.

1.5 Ethical and social implications of AIED

Despite the benefits of AIED, there are ethical and social handicaps, such as "data theft, privacy issues, digital divide, over control, and false outcomes" as well as "making learners less social" (Akgun & Greenhow, 2022, p.5). More importantly, "much AIED has been designed (whether intentionally or not) to supplant teachers or to reduce them to a functional role and not to assist them to teach more effectively" (Holmes et al., 2023, p.621). Thus, the rationale behind using AIED needs questioning and thus its function has to be repositioned in such a way to support teachers and learners in their effort to achieve quality education. All in all, "AI might help open up teaching and learning possibilities that are otherwise difficult to achieve, that challenge existing pedagogies, or that help teachers to be more effective" (Holmes et al., 2023, p.621).

Education is more than learning and instruction, and by nature it is human-to-human. That is, it is interactive and social. However, AI still runs the risk of damaging the social aspects and functions of education, popularizing the customization of education in the form of personalized AI tools. As Kohn (2015, p.14) rightly wrote, "…meaningful (and truly personal) learning never requires technology. Therefore, if an idea such as personalization is presented from the start as entailing software or a screen, we ought to be extremely skeptical about who really benefits." Then comes out the terrible face of wild capitalism for the sake of money-making at the expense of human development. Considering the fact that the strongly motivated students have the capacity to search for and find the information for their study needs, the request for personalized learning does not stem from learners' needs but from the demand by companies which aim to sell more software (Chassignol et al., 2018). Moreover, it is suggested that AI tools and application do not originate from universities but from profit organizations, such as LinkedIn, lynda.com, Amazon or Coursera, working with large data sets. At this point, Bates et al. (2020) rightly argue:

> "…this would pose an existential threat to public schools, colleges and universities. The issue then becomes: who is best placed to protect and sustain the individual in a digital age: multinational corporations or a public education system? The key question then is whether technology should aim to replace teachers and instructors through automation, or whether technology should be used to empower not only teachers but also learners. Above all, who should control AI in education: educators, students, computer scientists, or

large corporations? These are indeed existential questions if AI does become immensely successful in reducing the costs of teaching and learning: but at what cost to us as humans? Fortunately, AI is not yet in a position to provide such a threat, but this will not always be the case. The tsunami is coming" (Bates et al., 2020, p.11–12).

Thus, it is clear that the above concerns compel us to reconsider the way we apply AI to education. Moreover, using AI systems in education causes changes in the learning environments of students (e.g., Smart School Emergency Tents mentioned above). The person-in-environment approach in social work underlines the significance of considering an individual and individual behavior with respect to the social contexts in which that person leads their life and interacts with other people (Kondrat, 2013). This means any change in a given environment affects the individuals living in that environment. Therefore, we should consider the spillover effects of AI tools when we introduce them in education.

A further noteworthy point to remember is the significance of critical thinking skills in education. It is stated that lessons need to be constructed to promote critical thinking and problem solving through challenging activities that tap into analytical reasoning. Accordingly, AIED should be used in such a way as to provide opportunities for learners to use their critical thinking skills so that they can employ their technical knowledge in creative ways (Abulibdeh et al., 2024). For example, "AI systems, with their vast databases and analytical capabilities, can present students with complex problems and scenarios that require more than just rote memorization or basic understanding. These systems can challenge students to use higher-order thinking skills, such as analysis, synthesis, and evaluation, to navigate through these problems" (Walter, 2024, p.20). Further, as students' knowledge about AI increases and as they learn what they can and cannot do with AI systems, they can start questioning the technology, which offers them a good opportunity to sharpen their critical skills (Walter, 2024).

It is deemed essential by researchers and practitioners to have the scope, trends, and future of AIED understood. However, the vast amount of research accumulated over the years makes it challenging to gain a comprehensive overview. To address this, automated analyses play a crucial role in identifying popular topics and how they have evolved throughout the course of time (Blei & Lafferty, 2007). Since researchers cannot feasibly read the ever-increasing literature on themes in education, topic modeling algorithms offer valuable statistical methods to follow the developments in AIED. The terms in the articles, the contained themes, the connections between these themes, and their evolution over time are analyzed by topic modeling algorithms (Blei, 2012). These algorithms analyze the terms, themes, and connections within articles, unveiling hidden semantic issues and facilitating efficient analysis of extensive research (Blei, 2012; Blei & Lafferty, 2007). Accordingly, using topic modelling, this study aims to help the researchers see the topics and trends in AIED throughout time, from 1967 to 2023. Existing literature tends to focus on specific periods or limited aspects of AIED, creating a gap that the current study aims to fill. The studies reviewing trends and changes in AIED are limited, which makes these reviews even more vital, as they offer a foundation for future research, guiding scholars to address the existing gaps in the field.

Although they reported significant results, the studies so far have not provided scholars with enough information to gain a comprehensive understanding of AIED. For example, Zhai et al. (2021) explored AI using content analysis to uncover the use of AIED with possible trends and threats. They examined 100 papers, including 63 empirical papers selected from the education and educational research category of Social Sciences Citation Index database from 2010 to 2020. Emphasizing four research trends (namely, Internet of things, swarm intelligence, deep learning, and neuroscience) to further explore, they reported that the challenges in education could stem from inappropriate use of AI applications, techniques, changing roles of teachers and students, in addition to social and ethical concerns. In addition, Paek and Kim (2021, p.10) investigated the existing impact of AI on education. Using topic modelling, they analyzed the studies on AIED published in the last 20 years, since 2001. They found eight topics by "reviewing the coherence score and using LDA algorithm," and they enumerated the topics using words' probability of appearance. The labels of the topics were Topic 1: Changes in the content of teaching; Topic 2: Feedback on assessment and evaluation; Topic 3: Enhancing interaction in online learning; Topic 4: Learning strategies and academic achievement; Topic 5: Language learning and literacy; Topic 6: AI-driven edu-tech; Topic 7: Learning experience and medical education; Topic 8: Machine learning algorithm. Then they categorized the topics as hot and cold in terms of their sign of rising or falling and their statistical significance (hot: rising / cold: falling). They concluded that there was a lot of work to do in AIED, and studies had just started. AIED concepts required accepted definitions. There was a need for AIED contents to be diversified along with varied methods of research (Paek & Kim, 2021, p.13–17). In his bibliometric study, Talan (2021) went through 2,686 papers on AIED published between 2001–2021, with a special focus on AI's bibliometric properties. He found the majority of the studies were performed in the USA and that AI, ITSs, machine learning, deep learning and higher education were among the words encountered more frequently. Finally, reviewing 1830 research articles on AIED between 2010 and 2019, Feng and Law (2021) carried out a network-based keyword analysis to paint a general picture of the research in the field. They discovered ITSs (2010–19) and massive open online courses as the two recurrent themes. They also found online learning, game-based learning, collaborative learning, assessment, affect, engagement, and learning design as principal educational issues emerging from the AIED research.

Unlike the fairly limited time periods of the previous studies, this study provides a more comprehensive overview of the topic of AIED spanning 56 years. Existing reviews have reported investigations into the impact of AI on learning within restricted timeframes, thereby neglecting to explore the broader trends in AI implementation in education. To address this gap in the literature, the present study conducts an analysis of all relevant research on AIED from 1967 to 2023, employing the topic modeling technique. This study covers a broad time span and provides a detailed account of the historical development of the field and how research topics have changed. The central aim is to uncover the evolving trends and developments in AI's role in education and offer valuable guidance to researchers and educators in the field. This research will not only review the existing literature but also identify trends in the field. In this context, the study aims to bring new perspectives to research in the field of AIED by shedding light on issues that have previously been under-addressed in the literature. It can also lead researchers to future research opportunities. This approach will go beyond the existing literature in the field, analyzing topics over a wide time span to better understand the future directions of AIED research.

2 Method

The study was designed to investigate four research questions within the domain of AIED studies covering the years from 1967 to 2023:

RQ 1: What are the descriptive characteristics that define AIED studies during the period from 1967 to 2023?
RQ 2: Which research topics have been prominent in AIED studies from 1967 to 2023?
RQ 3: How have the primary themes in AIED studies evolved over the span of 56 years?
RQ 4: What future trends can be anticipated in AIED studies?

To examine the descriptive characteristics (RQ 1) and evolving research patterns (RQ 2, RQ 3 and RQ 4) in AIED studies, the research methodology was carefully structured, encompassing the essential stages of descriptive analysis and topic modeling. The research process was broken down into four core sub-processes: (1) Data Collection, (2) Descriptive Analysis, (3) Topic Modeling (see Fig. 2), and (4) Interpretation and Reporting. Additionally, this study adheres to the specific steps outlined in the PRISMA 2020 Statement for conducting systematic reviews, ensuring a comprehensive and standardized approach.

2.1 Data collection

Regarding the data sources for our analysis, we utilized two prominent bibliographic electronic databases: Web of Science and Elsevier's Scopus. Initially, we conducted searches in both databases using identical search queries. Subsequently, we opted for Elsevier's Scopus due to its broader coverage of journals and more effective search options than Web of Science (Mongeon & Paul-Hus, 2016).

The selection of appropriate search terms is a pivotal step. Hence, we employed an iterative process to identify effective search terms systematically. We initiated the search by using the term "Artificial Intelligence", and expanded with widely recognized terms such as "Machine Learning", "Deep Learning", "Expert System", "Machine Intelligence", "Neural Network", "Intelligent System", "Autonomous System", and "Cognitive Computing" in abstract, title, or keywords. Related terms were searched in sources that contained "education", "learn*", "teach*", or "instruction" in their title, abstract, or keywords. The asterisk symbol (*) was used to broaden the

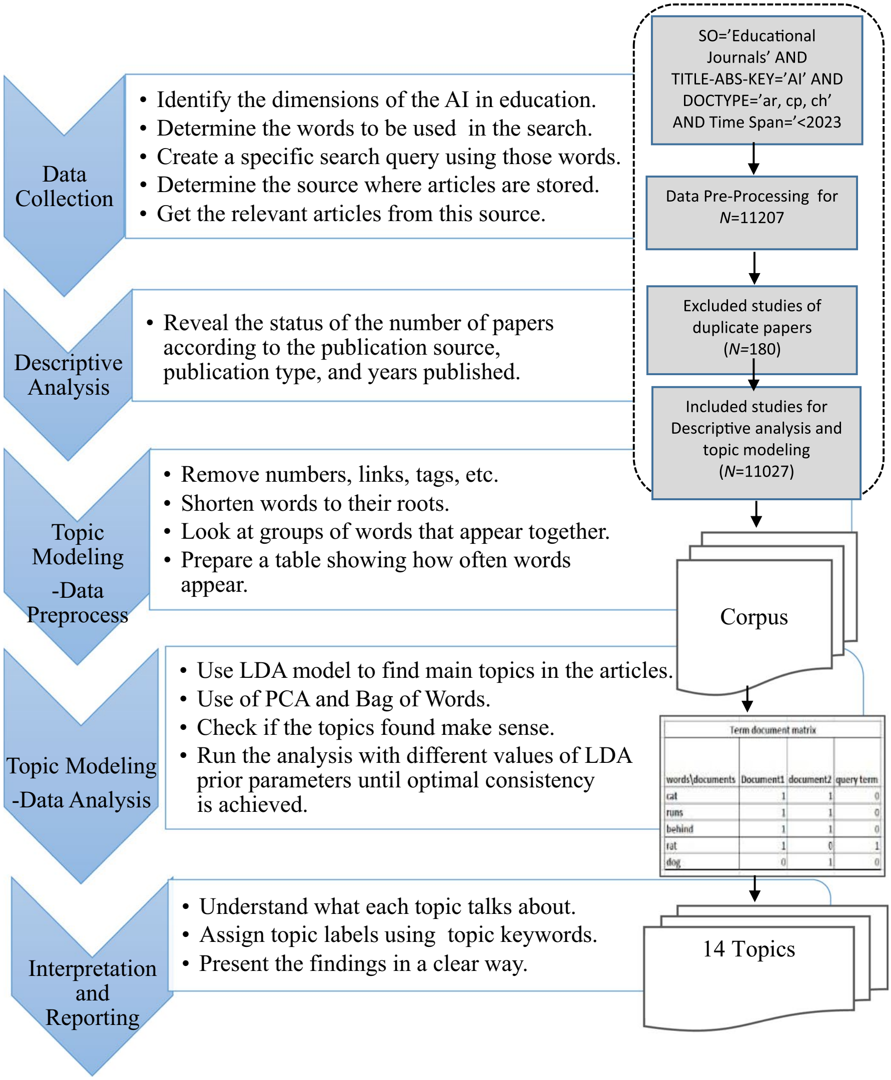


Fig. 2 Steps of the research methodology

search to include different variations or endings of the root word. The search criteria did not focus on a specific education context, such as primary, secondary, or higher education, hence it encompassed general keywords related to educational contexts. In topic modeling and descriptive analysis, title, abstract, and keywords are the primary material of the analysis. Thus, related terms were searched in these sections of the publications. By examining the results, sources related to computer science not related to education were removed from search results.

The final version of the search query after the iterative process looked like this:

"TITLE-ABS-KEY ("artificial intelligence" OR "ai" OR "machine learning" OR "deep learning" OR "expert system" OR "Neural Network" OR "intelligent system" OR "machine intelligence" OR "Autonomous System" OR "Cognitive Computing") AND SRCTITLE ("education" OR "learn*" OR "teach*" OR "instruction") AND (DOCTYPE (ar) OR DOCTYPE (cp) OR DOCTYPE (ch)) AND LANGUAGE (english) AND PUBSTAGE (final) AND NOT SRCTITLE ("deep learning" OR "machine learning" OR "Learning Representations" OR "Reinforcement Learning")"

In total, 11,207 journal publications in English were retrieved from Scopus published before July 2023. After excluding publications without an abstract or with duplicate abstracts, 11,027 publications were included as a final dataset for descriptive analysis and topic modelling.

2.2 Descriptive analysis

Descriptive analysis serves as the foundational step in comprehending and interpreting datasets, offering a structured framework for understanding fundamental characteristics and trends. In our study's descriptive analysis phase, articles were systematically analyzed based on various characteristics, including the distribution across subjects, years, countries, scientific contributions, influential authors, and publications (Donthu et al., 2021; Huang et al., 2020). This comprehensive approach aimed to unravel AI-related educational research by examining the distribution of papers across publication sources, types, and publication years.

2.3 Topic modeling

Topic modeling is a text analysis technique utilized in the field of natural language processing (NLP) and information retrieval. It is designed to uncover latent thematic structures within a collection of documents, enabling the identification of underlying topics or themes present in the vast volumes of textual data (Blei et al., 2003). This unsupervised machine learning method facilitates the extraction of valuable insights, document categorization, and information retrieval within large textual datasets. One of the most widely used algorithms for topic modeling is Latent Dirichlet Allocation (LDA) (Jelodar et al., 2019). LDA is employed to discover hidden themes within a document collection and determine the distribution of these themes within the documents. LDA treats each document as a composite of various topics and each topic as a mix of keywords, each in a specific proportion (Blei et al., 2003). It calculates the prevalence of topics within documents and the significance of words within those topics, all based on the distribution of words across the entire dataset. This process employs iterations of the Dirichlet distribution to effectively discern hidden semantic structures.

The LDA model associates text documents with a set of topics, identifies the words that define each topic, and quantifies the presence of these topics within each document. This is useful in applications such as automatic categorization, text mining, and information extraction, especially when dealing with large text collections.

LDA serves as a powerful tool for making text data more understandable and manageable, as well as for discovering latent themes and analyzing document groups.

2.4 Text cleaning and pre-processing

In the preparation of text data for LDA analysis, employing standard text cleaning methods is essential, particularly since the text data can be disordered, and topic modeling requires well-defined numeric inputs in vector format for the LDA algorithm (Blei et al., 2003). This transformation of text into a vector of numbers is crucial for LDA, ensuring that the algorithm can effectively uncover latent topics while preserving the linguistic and semantic properties of the text. The text cleaning and pre-processing steps applied to facilitate the use of text data in LDA-based topic modeling are outlined below:

1. Importing the text data into the Python environment.
2. Breaking down the text file into individual sentences.
3. Segmenting each sentence into its constituent words.
4. Ensuring uniformity by converting all words to lowercase.
5. Eliminating punctuation from each word to focus on the core text.
6. Excluding words that do not consist of alphabetic characters.
7. Removing common stop words (e.g., "the," "and," "is") that may not contribute significantly to topic modeling.
8. Further refining the list of stop words to exclude those not relevant to LDA analysis.

Following text cleaning and pre-processing, the subsequent research methods were employed:

1. Counting word frequencies and sentence frequencies.
2. Generating a word cloud.
3. Conducting topic modeling using the LDA technique with Bag of Words as the text vector (Zhang et al., 2010).
4. Creating an inter-topic distance map using principal component analysis (PCA).
5. Constructing a text network.

The visual representation of these processes is illustrated in Fig. 2.

2.5 Interpretation and reporting

Once topic modeling is applied to a corpus of text, it is essential to interpret the results effectively. This involves understanding what each discovered topic represents by analyzing the top words (i.e., mentioned most frequently) associated with each topic, thereby discerning the underlying theme or subject matter that these words collectively convey. To enhance accessibility and interpretability, labels were assigned to topics based on the keywords and the context of the top words within each topic. These labels succinctly describe the core subject matter of each topic. Additionally, findings were presented in a clear and organized manner by creating visualizations such as word clouds, bar charts, and tables to illustrate the distribution of topics and their keywords within the text corpus. Furthermore, a summary or brief description of each topic was provided to facilitate understanding.

3 Results

3.1 Descriptive analysis result (RQ 1)

The descriptive analysis focused on the distribution of articles concerning five specific aspects: year of publication, publisher, publication type, number of publications in countries, and subject areas in response to Research Question 1 (RQ 1). While the data in Table 1 presents a breakdown of the number of publications categorized within ten-year intervals, Fig. 3 shows the number of publications by year. The earliest recorded studies indexed in the SCOPUS database date back to 1967, marking the inception of this academic field. Subsequently, the field experienced a gradual increase in scholarly output during the period from 1967 to 2003, accounting for 772 publications (6.9%), and there is a sharp increase, especially after 2004. While the years between 2004 and 2013 cover 20.3%, with 522 publications, the total number of publications between 2014 and 2023 increased by approximately three times and covered 72.6%.

Table 2 presents an examination of the top 15 publication sources contributing to the analyzed studies, out of a total of 11,027 studies. "Proceedings—Frontiers in Education Conference" represent the largest share at 15% ($n=328$), with "International Journal of Emerging Technologies in Learning" closely following with 248 studies, constituting 11% of the total. Additionally, it is worth noting that two of the top three publication sources are specifically associated with conferences.

Figure 4 offers insights into the categorization of publication types. Among the total publications, 55% were conference papers ($n=6100$), followed by articles at 38% ($n=4196$), and book chapters at 7% ($n=731$). These results are consistent with the search profile, as would be expected.

Figure 5 displays the publication counts for the top 15 countries with the highest number of publications, out of a total of 11,027 publications. The United States ($n=2512$) and China ($n=2260$) are the top two countries with the highest

Table 1 Distribution of publications in ten-year periods

Yearly periods	n	%
2023–2014	8010	72.64%
2013–2004	2241	20.32%
2003–1994	522	4.73%
1993–1984	237	2.15%
1983–1974	13	0.12%
1973–1967	4	0.04%
Total	11,027	100.00%

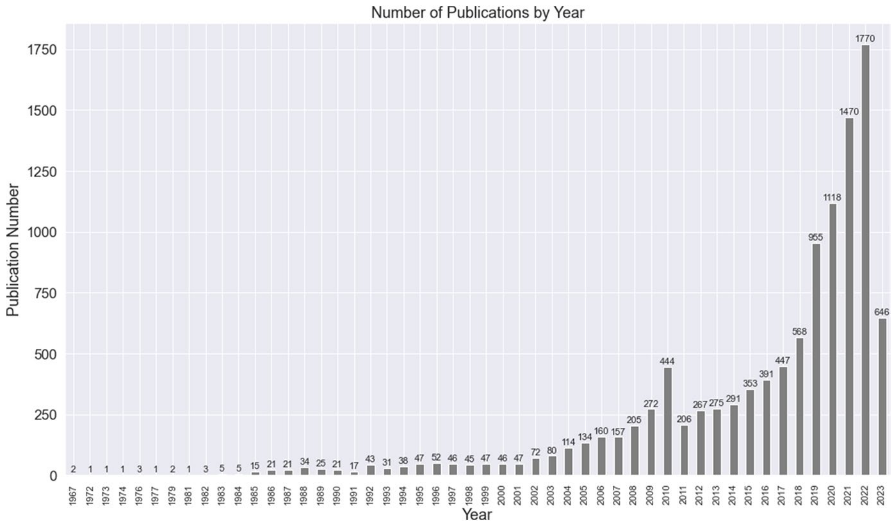


Fig. 3 Number of publications by year. Note. 2023 covers publications until July 2023

publication outputs. In contrast, Singapore ($n = 154$) and Turkey ($n = 155$) are the countries with the least number of publications in this field.

Table 3 presents information about the subject areas encompassing the publications within the dataset across 15 different fields. The majority of the publications are from Social Science ($n = 9575$) and Computer Science ($n = 7292$). It is important to acknowledge that certain articles may belong to multiple subject areas, indicating interdisciplinary research trends within the dataset.

3.2 LDA analysis (topic modeling) results (RQ2)

To address RQ 2, topic analysis was conducted using an LDA model. Hence, 14 distinct topics were identified for the 11,027 articles. Table 4 presents the topic keywords and the corresponding topic labels that were determined primarily using the top five keywords associated with each topic. The ranking presented in Table 4 is based on the number of articles associated with each topic. Therefore, topics with the highest number of articles are listed at the top. The top-rated topics are as follows: Topic 2, labeled as "AI for Engineering Students" ($n = 1287$, 11.7%); Topic 13, labeled as "Neural Networks and Image Recognition" ($n = 1266$, 11.5%); and Topic 3, labeled as "Research Review in Educational Study" ($n = 1143$, 10.4%) (Table 4).

3.3 Results of volume analysis of each topic (RQ3)

In order to identify the topic trends over time, the number of articles for each topic was examined for every 10-year period, and the top three topics with the

Table 2 Publication sources

Source name	# of papers	%
Proceedings—Frontiers in Education Conference, FIE	328	15%
International Journal of Emerging Technologies in Learning	248	11%
Proceedings—2010 International Conference on Artificial Intelligence and Education, ICAIE, 2010	195	9%
Computers and Education	189	9%
IEEE Global Engineering Education Conference, EDUCON	162	7%
Proceedings of the International Conference on Education and Research in Computer Aided Architectural Design in Europe	152	7%
Education and Information Technologies	152	7%
International Journal of Artificial Intelligence in Education	129	6%
International Journal of Engineering Education	102	5%
Computer Applications in Engineering Education	99	5%
Computers and Education: Artificial Intelligence	95	4%
IEEE Transactions on Learning Technologies	89	4%
IEEE Transactions on Education	80	4%
Proceedings of 2022 IEEE 11th Data Driven Control and Learning Systems Conference, DDCLS 2022	77	4%
Proceedings of 2021 IEEE 10th Data Driven Control and Learning Systems Conference, DDCLS 2021	71	3%
Total	2168	100%

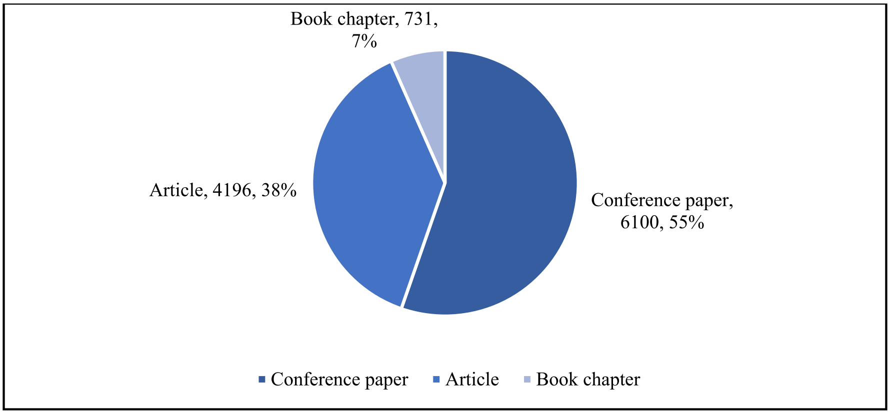


Fig. 4 Distribution of the publications

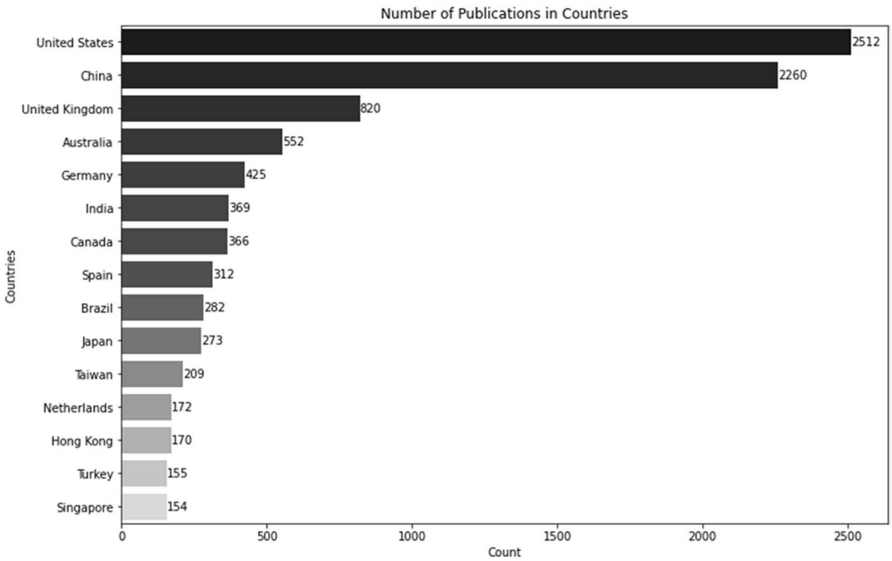


Fig. 5 Number of publications in countries

highest article counts during each period were determined sequentially. The relevant topics are presented in Table 5. Except for the period 2014–2023 and 1973–1967, "AI for Engineering Students" consistently emerged as the top-rated topic in all other 10-year periods.

Table 6 contains specific information in its third column, representing the percentage of articles that each topic holds within a particular 10-year period, relative to all topics combined. For example, during the period from 1984 to 1993, the topic "AI for Engineering Students" accounted for 53.6% of all articles published

Table 3 Subject areas of the publications

Subject areas	Number of results
Social Sciences	9575
Computer Science	7292
Engineering	3334
Decision Sciences	1147
Mathematics	973
Business, Management and Accounting	544
Psychology	520
Medicine	442
Arts and Humanities	254
Neuroscience	253
Energy	185
Health Professions	158
Nursing	89
Agricultural and Biological Sciences	74
Physics and Astronomy	58

in that decade among the 14 topics. The fourth column indicates the acceleration value (Δ), representing the increase or decrease in the number of articles compared to the previous 10-year period. The acceleration value is calculated by taking the difference in the number of articles between the current decade and the previous decade and by dividing it by 10. In the fifth column, the number of articles associated with the respective topic for each 10-year period is presented. In the sixth and seventh columns, there are graphs showing the change of the data given in the third and fourth columns in 6 periods of 10 years. Specifically, "AI for Engineering Students" constitutes 26.1% of related studies across all decades, followed by “Student Performance Assessment” (18.1%), “Neural Network Algorithms in Robotics and Detection” (12.2%), “Language Processing and Problem Solving” (6.3%), and “Neural Networks and Image Recognition” (5.1%).

3.4 Result of acceleration analysis (RQ 4)

To better comprehend and interpret the future trends in the utilization of AI in educational research (RQ 4), acceleration values (Δ) were employed for each topic within each decade. In Table 6, the acceleration values (Δ) for each topic are evident. These values were calculated by considering changes in the number of articles between the current decade and the preceding one. In Fig. 6, the average acceleration value for each topic is provided. As observed in Fig. 6, topics such as "AI for Engineering Students", "Performance Assessment", "Neural Network Algorithms in Robotics and Detection", "Language Processing and Problem Solving", and "Deep Learning Approaches in Teaching" exhibit a very fast pace when considering changes in the number of publications (greater than 3.17). Conversely, topics such as "AI in Medical Education" and "Digital Technology"

Table 4 LDA topics and yearly distribution of topics

Topic label	Topic keywords	2023–2014	2013–2004	2003–1994	1993–1984	1983–1974	1973–1967	# of papers	%
Topic 2: AI for Engineering Students	student, engineering, system, computer, course, project, science, software, program, intelligence, artificial, paper, development, curriculum, skill	653	324	177	127	6	0	**1287**	**11.67**
Topic 13: Neural Networks and Image Recognition	network, neural, model, method, image, feature, deep, recognition, based, result, classification, proposed, using, paper, detection	1035	190	34	7	0	0	**1266**	**11.48**
Topic 3: Research Review in Educational Study	education, research, study, social, educational, higher, development, review, work, experience, paper, technology, community, context, analysis	952	157	26	8	0	0	**1143**	**10.37**
Topic 1: Student Performance Assessment	student, study, approach, performance, assessment, result, group, academic, deep, effect, factor, level, using, score, test	699	200	37	7	4	2	**949**	**8.61**
Topic 6: Machine Learning in Data and Performance Prediction	data, model, machine, prediction, performance, algorithm, using, method, used, accuracy, result, classification, technique, analysis, mining	788	130	14	1	0	0	**933**	**8.46**
Topic 10: Online Course and Feedback	student, course, online, feedback, class, active, classroom, activity, instructor, lecture, engagement, result, using, tool, video	742	147	22	3	0	0	**914**	**8.29**

Table 4 (continued)

Topic label	Topic keywords	2023–2014	2013–2004	2003–1994	1993–1984	1983–1974	1973–1967	# of papers	%
Topic 4: Language Processing and Problem Solving	language, solve, problem, question, task, process, machine, human, approach, agent, using, programming, natural, knowledge, word	570	226	56	23	0	0	**875**	**7.94**
Topic 11: Deep Learning Approaches in Teaching	teaching, student, learning, approach, deep, education, concept, practice, skill, thinking, classroom, knowledge, activity, understanding, new	531	226	47	10	1	0	**815**	**7.39**
Topic 12: Educational AI Technology	teaching, artificial, education, technology, intelligence, based, data, information, management, evaluation, research, development, paper, system, method	617	143	2	2	1	0	**765**	**6.94**
Topic 7: Neural Network Algorithms in Robotics and Detection	system, network, control, algorithm, neural, based, method, robot, using, proposed, paper, function, simulation, detection, result	514	187	34	4	0	2	**741**	**6.72**
Topic 8: Adaptive E-Learning and Personalized Education	system, learner, user, e-learning, knowledge, personalized, based, educational, adaptive, intelligent information, environment, online, web, model	228	147	19	5	0	0	**399**	**3.62**

Table 4 (continued)

Topic label	Topic keywords	2023–2014	2013–2004	2003–1994	1993–1984	1983–1974	1973–1967	# of papers	%
Topic 5: Cognitive Simulation and Computational Design	design, process, model, cognitive, simulation, knowledge, building, tool, theory, expert, method, decision, computational, architecture, fuzzy	162	106	37	28	0	0	**333**	**3.02**
Topic 14: New Digital Technology/Smart Education	technology, digital, new, machine, data, intelligence, artificial, child, industry, smart, education, Covid19, world, application, need	303	18	5	4	1	0	**331**	**3.00**
Topic 9: AI in Medical Education	ai, medical, artificial, intelligence, health, patient, clinical, training, education, school, literacy, care, program, participant, curriculum	216	40	12	8	0	0	**276**	**2.50**
Total		**8010**	**2241**	**522**	**237**	**13**	**4**	**0**	**100**

Table 5 Top three trend topics by decade

Year range	Topic/s 1	Topic/s 2	Topic/s 3
2023–2014	Neural Networks and Image Recognition	Research Review in Educational Study	Machine Learning in Data and Performance Prediction
2013–2004	AI for Engineering Students	Deep Learning Approaches in Teaching **&** Language Processing and Problem Solving	Student Performance Assessment
2003–1994	AI for Engineering Students	Language Processing and Problem Solving	Deep Learning Approaches in Teaching
1993–1984	AI for Engineering Students	Cognitive Simulation and Computational Design	Language Processing and Problem Solving
1983–1974	AI for Engineering Students	Student Performance Assessment	Deep Learning Approaches in Teaching **&** Deep Learning Approaches in Teaching **&** New Digital Technology
1973–1967	Student Performance Assessment **&** Neural Network Algorithms in Robotics and Detection	Control and Neural Network Algorithms in Robotics and Detection Systems	

Table 6 Mean and acceleration values (Δ) of topic according to decade

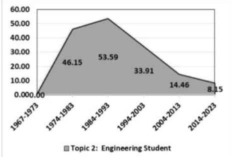
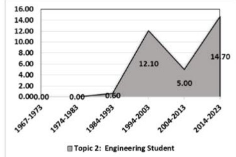
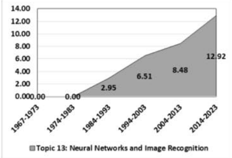
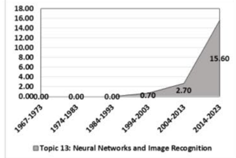
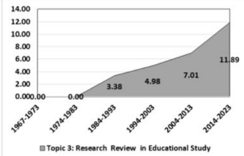
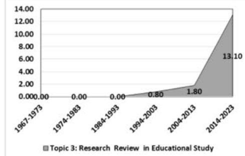
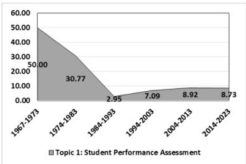
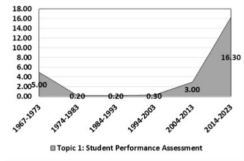
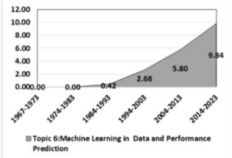
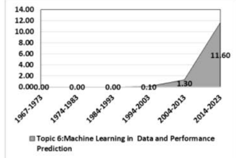
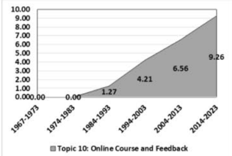
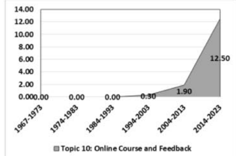
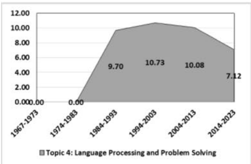
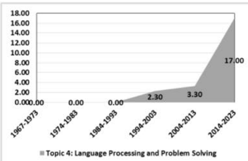
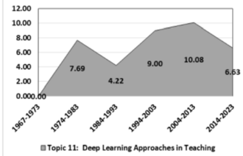
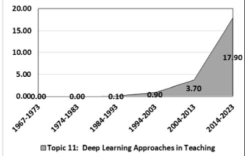
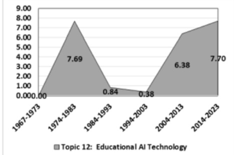
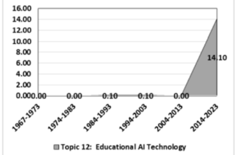
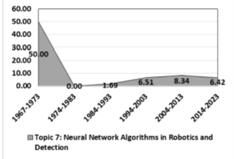
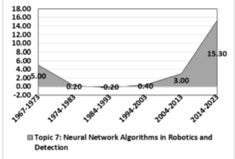
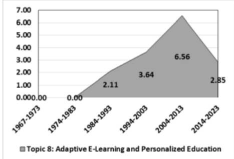
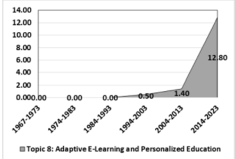
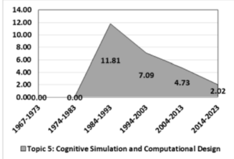
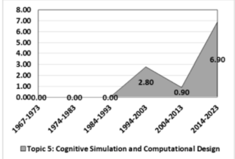
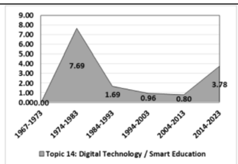
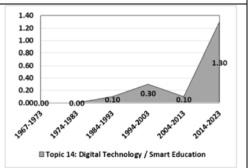
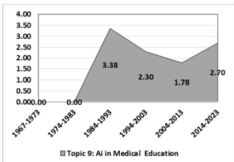
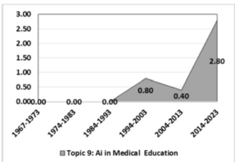
Topic Label	Year Range	%	Δ	N	Mean Volume Graph	Acceleration Graph
AI for Engineering Students	1967-73	0.0	0.0	0		
	1974-83	46.2	0.0	6		
	1984-93	53.6	0.6	127		
	1994-03	33.9	12.1	177		
	2004-13	14.5	5.0	324		
	2014-23	8.2	14.7	653		
	AVG	26.1	5.4	214.5		
Neural Networks and Image Recognition	1967-73	0.0	0.0	0		
	1974-83	0.0	0.0	0		
	1984-93	3.0	0.0	7		
	1994-03	6.5	0.7	34		
	2004-13	8.5	2.7	190		
	2014-23	12.9	15.6	1035		
	AVG	5.1	3.2	211		
Research Review in Educational Study	1967-73	0.0	0.0	0		
	1974-83	0.0	0.0	0		
	1984-93	3.4	0.0	8		
	1994-03	5.0	0.8	26		
	2004-13	7.0	1.8	157		
	2014-23	11.9	13.1	952		
	AVG	4.5	2.6	190.5		
Student Performance Assessment	1967-73	50.0	5.0	2		
	1974-83	30.8	0.2	4		
	1984-93	3.0	0.2	7		
	1994-03	7.1	0.3	37		
	2004-13	8.9	3.0	200		
	2014-23	8.7	16.3	699		
	AVG	18.1	4.2	158.2		
Machine Learning in Data and Performance Prediction	1967-73	0.0	0.0	0		
	1974-83	0.0	0.0	0		
	1984-93	0.4	0.0	1		
	1994-03	2.7	0.1	14		
	2004-13	5.8	1.3	130		
	2014-23	9.8	11.6	788		
	AVG	3.1	2.2	155.5		
Online Course and Feedback	1967-73	0.0	0.0	0		
	1974-83	0.0	0.0	0		
	1984-93	1.3	0.0	3		
	1994-03	4.2	0.3	22		
	2004-13	5.6	1.9	147		
	2014-23	9.3	12.5	742		
	AVG	3.4	2.5	152.3		
Language Processing and Problem Solving	1967-73	0.0	0.0	0.0		
	1974-83	0.0	0.0	0.0		
	1984-93	9.7	0.0	23.0		
	1994-03	10.7	2.3	56.0		
	2004-13	10.1	3.3	226.0		
	2014-23	7.1	17.0	570.0		
	AVG	6.3	3.8	145.8		

Table 6 (continued)

Deep Learning Approaches in Teaching	1967-73	0.0	0.0	0.0		
	1974-83	7.7	0.0	1.0		
	1984-93	4.2	0.1	10.0		
	1994-03	9.0	0.9	47.0		
	2004-13	10.1	3.7	226.0		
	2014-23	6.6	17.9	531.0		
	AVG	6.3	3.8	135.8		
Educational AI Technology	1967-73	0.0	0.0	0.0		
	1974-83	7.7	0.0	1.0		
	1984-93	0.8	0.1	2.0		
	1994-03	0.4	0.1	2.0		
	2004-13	6.4	0.0	143.0		
	2014-23	7.7	14.1	617.0		
	AVG	3.8	2.4	127.5		
Neural Network Algorithms in Robotics and Detection	1967-73	50.0	5.0	2.0		
	1974-83	0.0	0.2	0.0		
	1984-93	1.7	0.2	4.0		
	1994-03	6.5	0.4	34.0		
	2004-13	8.3	3.0	187.0		
	2014-23	6.4	15.3	514.0		
	AVG	12.2	4.0	123.5		
Adaptive E-Learning and Personalized Education	1967-73	0.0	0.0	0.0		
	1974-83	0.0	0.0	0.0		
	1984-93	2.1	0.0	5.0		
	1994-03	3.6	0.5	19.0		
	2004-13	6.6	1.4	147.0		
	2014-23	2.9	12.8	228.0		
	AVG	2.5	2.5	66.5		
Cognitive Simulation and Computational Design	1967-73	0.0	0.0	0.0		
	1974-83	0.0	0.0	0.0		
	1984-93	11.8	0.0	28.0		
	1994-03	7.1	2.8	37.0		
	2004-13	4.7	0.9	106.0		
	2014-23	2.0	6.9	162.0		
	AVG	4.3	1.8	55.5		
New Digital Technology/Smart Education	1967-73	0.0	0.0	0.0		
	1974-83	7.7	0.0	1.0		
	1984-93	1.7	0.1	4.0		
	1994-03	1.0	0.3	5.0		
	2004-13	0.8	0.1	18.0		
	2014-23	3.8	1.3	303.0		
	AVG	2.5	0.3	55.2		
AI in Medical Education	1967-73	0.0	0.0	0.0		
	1974-83	0.0	0.0	0.0		
	1984-93	3.4	0.0	8.0		
	1994-03	2.3	0.8	12.0		
	2004-13	1.8	0.4	40.0		
	2014-23	2.7	2.8	216.0		
		1.7	0.7	46.0		
	AVG					

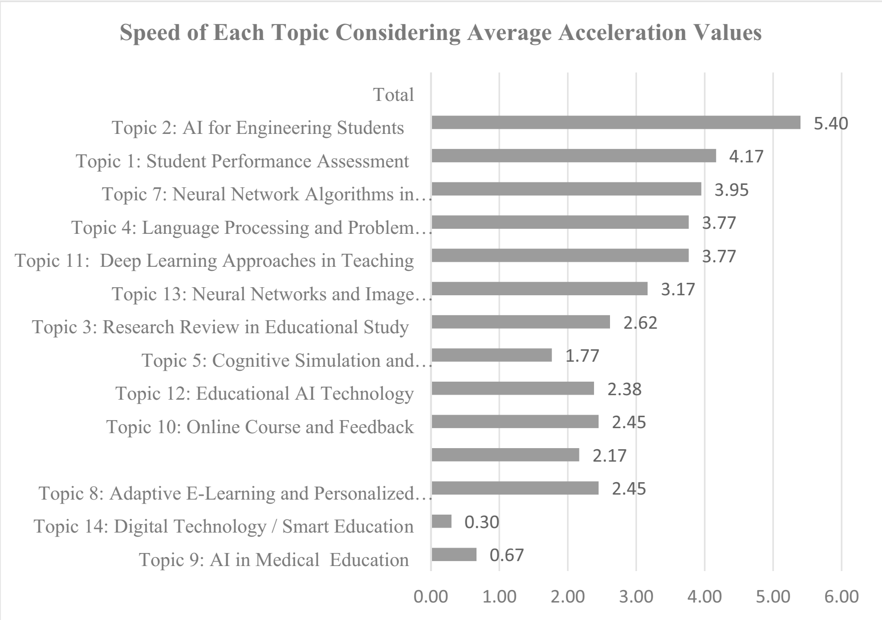


Fig. 6 Acceleration speed of topics

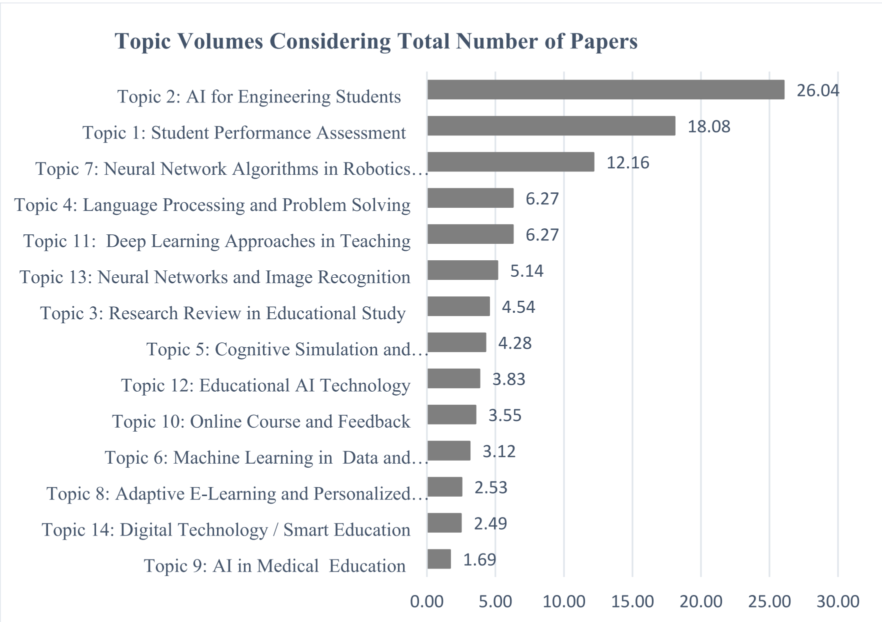


Fig. 7 Average percentage of topics based on the count of publications per decade

are comparatively slower (less than 1.77), while the remaining topics fall within the medium pace range (between 1.77 and 3.17). Figure 7 presents an analysis based on the average percentage of topics based on the count of publications per decade. The topic "AI for Engineering Students" has the highest volume, comprising 26.04% of the total papers, indicating a significant focus in this area. Following this, "Student Performance Assessment" accounts for 18.08% of the papers, suggesting a substantial interest in evaluating student performance using AI technologies.

4 Discussion

The findings have been grouped and discussed under the following five headings: ***Trends and Acceleration in AIED Research (RQ1, RQ2, RQ3, RQ4)***, which covers the rapid increase in the number of AIED research and changes in accelerations, AIED publication trends with milestones in 2010 and 2020, the role of conferences and journals in AIED research dissemination, and the contributions of different countries, particularly the USA and China; ***Interdisciplinary Contributions and Publication Types (RQ1)***, which discusses the interdisciplinary nature of AIED research, with contributions from various fields such as educational sciences, engineering, and medical education, and analyzes the types of publications prevalent in AIED research; ***AI Applications in Education (RQ2, RQ3)***, which explores various applications of AI in education, including its use for engineering students as both a tool and a lesson, the impact of neural networks and image recognition on learning experiences, innovative approaches in education with deep learning for personalization and assessment, and the power of machine learning in predicting student performance and providing customized recommendations; ***Assessment and Monitoring in Education (RQ2, RQ3)***, which examines the role of AI in measuring student achievement, assessment, and monitoring, highlighting innovations in robotics and detection, and their implications for educational practices; and ***Challenges and Future Directions (RQ4)***, which addresses the challenges and future directions for AI in education, with a specific focus on the high costs and barriers to specialization in using AI in medical education.

4.1 Trends and acceleration in AIED research (RQ1, RQ2, RQ3, RQ4)

4.1.1 Rapid increase in the amount of AIED research and the changes in accelerations

With the spread of computer and internet technologies, AI has emerged as a solution for collecting and analyzing big data (Miao et al., 2021) in the field of education (Luan et al., 2020). With the Covid-19 pandemic, distance education has significantly increased at the K-16 level in many regions over the last four years (Miao et al., 2021). The frequent use of educational technologies, such as AI-based learning management systems (LMSs) (UNESCO, 2021), the examination of learning analytics (Miao et al., 2021), and the use of adaptive learning systems have made AI widely used. Consequently, studies in this field have accelerated since 2004, with most of the publications occurring in the last ten years (Tables 2 and 4), indicating an increase in AI studies. Research has concluded that AIED is a rapidly emerging field (Chen et al., 2020a; Zhai et al., 2021). Conducting a bibliometric study into AIED, Talan (2021) arrives at the same conclusion, noting that research interest in AI in education has increased.

On the whole, all of the topics identified in the current study have seen a significant increase in acceleration in the last decade. This may be due to the tremendous advances in AI technologies, innovations in education, or global research trends in this period. The period from 2010 onwards is called the "Deep Learning Era," when the language and image recognition capabilities of AI systems have become "comparable to" humans' (Roser, 2024, p.1). Specifically, among the topics of the present study, the highest acceleration (5.4) belongs to Topic 2 (AI for Engineering Students) (Fig. 6). The second highest topic (4.17) is Topic 1 (Student Performance Assessment). This finding supports previous literature. It is known that AI has been used in different types of educational assessments (Hrich et al., 2024, p.823). Further, using AI in student performance assessment has many benefits such as "improved efficiency, objectivity, and scalability" (Hrich et al., 2023, p.2120). Another use of AI in student performance assessment can be self-assessment. It was reported that AI-powered learning analytics could be used by students for self-assessment to monitor their academic performance (Demartini et al., 2024). As AI tools and applications develop further, it seems that they will reshape student performance assessment.

4.1.2 AIED publication trends: 2010 and 2020 milestones

The first publications are from 1967 and there are very few publications until 1985 (Fig. 5). From 1985 to 2010, there was an overall increasing acceleration in the number of publications. There was a steady increase in the number of publications in AIED in the period between 1985 and 2010, but that period seems to be less active than the period between 2010 and 2020. This can be explained by two reasons. The first is that the computer technologies were primitive compared to today's AI (e.g., Commodore announced the Amiga 1000 in 1985). The second reason is that computer technologies may not have been as popular in education as they were in the 2000s and onwards.

From 2010 onwards, despite some fluctuations, there has been an ascending trend line in AIED so far (Fig. 3). Certain developments may have caused the publication interests to increase in different aspects of education with varied AI applications. In 2010, there was a rapid increase in research in the field of AIED due to the developments in computing systems, increasing storage power, big data, the development of artificial neural networks, the development of open-source AI algorithms, and the use of AI by large companies such as Google, Amazon, Facebook, etc. Today, AI-based systems, such as AI-powered toys, are used in early childhood education (Su et al., 2023a). Further, AI started to be applied in different aspects and practices of education from 2010 onwards. As an example, mobile learning technologies based on AI systems have become more widespread in distance learning and e-learning platforms (Liu et al., 2010). Introduced in 2022, ChatGPT has aroused considerable interest and is among the most debated topics in education today (Adeshola, & Adepoju, 2023; Deng & Lin, 2022; Farhi et al., 2023). Important meetings and academic workshops started to be held in 2011 to attract attention to using AIED for supporting the inclusion of communities (Akhras & Brna, 2011). More importantly, more than two hundred articles were published as a result of the International Conference on Artificial Intelligence and Education (ICAIE) held on October 29–30, 2010 (ICAIE, 2010). Thus, as new AI tools and applications are created and start to be used in education, research follows these developments and new publications are added into the AIED literature.

4.1.3 AIED in the academic arena: Conferences or journals?

Computer science is relatively young and most of its publications are conference publications (Franceschet, 2010). Similar to other studies (Ekin et al., 2023; Franceschet, 2010), we found in this study that more than half of the publications in AIED are conference papers (55%), the smallest pecentage is book chapters (7%), and the remaining 38% are articles (Fig. 3). The similarity in the results seems to stem from AIED's position in between computer science and education.

Our study shows that scholars preferred attending conferences to writing book chapters and articles to share their opinions and research results. Similarly, the number of submissions to outstanding AI conferences has gradually increased recently (Leyton-Brown et al., 2024). For example, there were fewer than 1000 submissions to the AAAI Conference on Artificial Intelligence (AAAI) between 1980–2012 in each year, but in 2021, it received more than 9000 submissions (Leyton-Brown et al., 2024). Conferences offer some advantages for the academics. First, it is known that "academic conferences have been an important means of scholarly communication and the dissemination of tacit knowledge among researchers for decades" (Kulczycki et al., 2024, p.303). Second, they provide researchers coming from various places with the opportunities for "intensive and in-depth discussions", "academic advancement", "feedback from various perspectives", and "cultural understanding" (Park et al., 2024, p.137). Thus, researchers may have preferred conference proceedings to articles or book chapters because of these advantages.

In terms of the sources of publications and the number of publications from these sources, the highest number of publications belongs to the publications presented at the Frontiers in Education Conference (Table 2). While six of these are conferences, the remaining nine are journals. The journal with the highest number of publications is the International Journal of Emerging Technologies in Learning. Possible reasons for this are (1) it is one of the oldest international journals combining technology and learning (it was launched in 2006), (2) it mainly accepts articles about learning through technological tools (some of the fields of interests listed on the journal's website are Online Learning Tools, MOOCs, SPOCs (Small Private Online Courses), Semantic Web Technologies, MashUp Technologies, Platforms and Content Authoring, New Learning Models and Applications, Internet Applications, Networked

Tools, Collaboration Tools / Collaborative Networks), (3) its publication frequency is twice a month, (4) the publication process is not complicated (IJET, 2024).

4.1.4 The role of countries in AI research: USA and China

It is known that the USA and China are the top research publishers in the world, in almost every knowledge field (Scimagojr, 2024). We found in our study that the USA and China are far ahead of other countries in terms of researchers' interest in AIED with approximately half of the publications belonging to these two countries (Fig. 4). It seems that the USA is progressively followed by other countries (such as China, in our study). In other studies, the USA was found as the country with the highest number of publications in computer sciences and AIED (Franceschet, 2010; Hinojo-Lucena et al., 2019; Roll & Wylie, 2016).

The fact that the USA was the leading spender of the top 10 countries with companies that spent the most on research and development (R&D) in 2022 (Long, 2024) and that China ($422.1 billion in 2022 according to The State Council of China) made extensive investments in R&D and encouraged the use of new technologies (China Daily, 2023) could be the driving force for academics to conduct research on these issues. For instance, Wang and Li (2023) indicated that China had foreseen the importance of AI well before 2015 and had progressively taken supporting steps related to AI for the development of AI systems. In addition, the fact that funding organizations encourage research on AI may account for the increase in these numbers. A final point to consider is that the USA and China are among the countries that have developed a national AI strategy (Pedro et al., 2019).

4.2 Interdisciplinary contributions and publication types (RQ1)

4.2.1 AIED in an interdisciplinary perspective: Contributions of different fields

Educational sciences are classified as a subfield of social science, and thus social sciences as a field of research discipline is the most active in the topic of AIED (Table 3). AI is studied by many different disciplines, from medical sciences to social sciences (Hernández-Lugo, 2024; Miller, 2019; Zheltukhina et al., 2024). For instance, *Nursing Education Today* has a special section to publish papers on technology-related issues in nursing education, including AI-based applications such as AI-chatbots (Tam et al., 2023). Thus, it can be expected that AI tools, applications, and their usefulness will be subjects of prospective research in different fields of science.

4.2.2 Types of publications

The third trending topic throughout the period studied is Research Review in Educational Study (Topic 3) (Table 4). In addition, the analysis by decade has shown that the number of studies in the last decade has increased significantly for Research Review in Educational Study (Topic 3). Further, Topic 3 is ranked the second out of the three trending topics in the last decade (2023–2014) (Table 5). As Eynon (2024) indicates, AIED deals with both basic and applied forms of research, and developers design systems to meet educational challenges. Educational problems and technological solutions that help us to eliminate these problems are increasing day by day, which makes the area of AIED lively and a rapidly-developing area of research. Due to this feature of AIED, reviews are needed to identify and sometimes to change the direction of research. This can be one reason why Topic 3 has become a trending topic. Further, "extensive review articles play an important role as a foundation for all types of research and its application" (Bahishti, 2021, p.1). Exploring how AIED evolves and predicting the prospective opportunities for AIED requires collecting, and sometimes combining, the results of the research in AIED by explaining the type and focus of the study, the specific subdomain and breadth, interaction type and collaborative structure, technology used, learning setting, and learning goals (as in the study by Guo et al., 2024). As a result, it is not unexpected to find that Research Review in Educational Study has become a trending topic.

4.3 AI applications in education (RQ2, RQ3)

4.3.1 AI for engineering students: AI as both a tool and a course

The most trending topic of the past 56 years seems to be Topic 2 (AI for Engineering Students) (Table 4). As in our study, engineering students are among the most popular study groups for researchers exploring AIED (Deo et al., 2020; Jiao et al., 2022; Ouyang et al., 2023; Rathi & Rathi, 2019). In addition, technical and analytical skills are central in engineering practices (Fleming et al., 2024). As both AI and machine learning require both skills, it is typical for researchers to focus on this area, considering the fact that "strong analytical ability, [the ability] to adapt and learn new technology, problem-solving within stipulated period[s] or deadlines, and work promotion are the most important factors for a successful future job performance" for engineering students (Gope & Gope, 2022, p.173). Further, AI-based educational technologies may be employed in the education of engineering students to enhance their learning experience, similar to those performed with the integration of robots in engineering education curriculum (Cardeira et al., 2006). Additionally, the use of simulation is important in engineering education and AI-based simulations provide environments that are more realistic. For example, virtual reality (VR) offers alternatives to the existing technologies utilized in engineering education, and computational fluid dynamics (CFD) simulations are the tool frequently used in designing and analyzing chemical engineering problems (Solmaz et al., 2024). Another issue is that since engineering education includes the teaching of new technologies, AIED is both a current and a future topic in engineering education. Moreover, one purpose of teaching AI in engineering education is that AI has become important for the industry today. In particular, prospective engineers are supposed to take action in complicated situations and create original technological and sustainable choices of solutions. To succeed in this, thus, they need to show "a T-shaped profile, where the vertical bar represents depth and mastery in one or more technical fields or disciplines, while the horizontal bar represents broader professional skills to address global, societal, and multidisciplinary challenges, such as thinking critically, working collaboratively, and communicating ideas" (Doulougeri et al., 2024, p.2). Finally, the current study found that computer science was among the main fields that constituted the majority of the publications by the USA and China. In short, both the use of AI in the education of engineers and the fact that they receive AI education may have played a role in making this Topic 2 a trending topic.

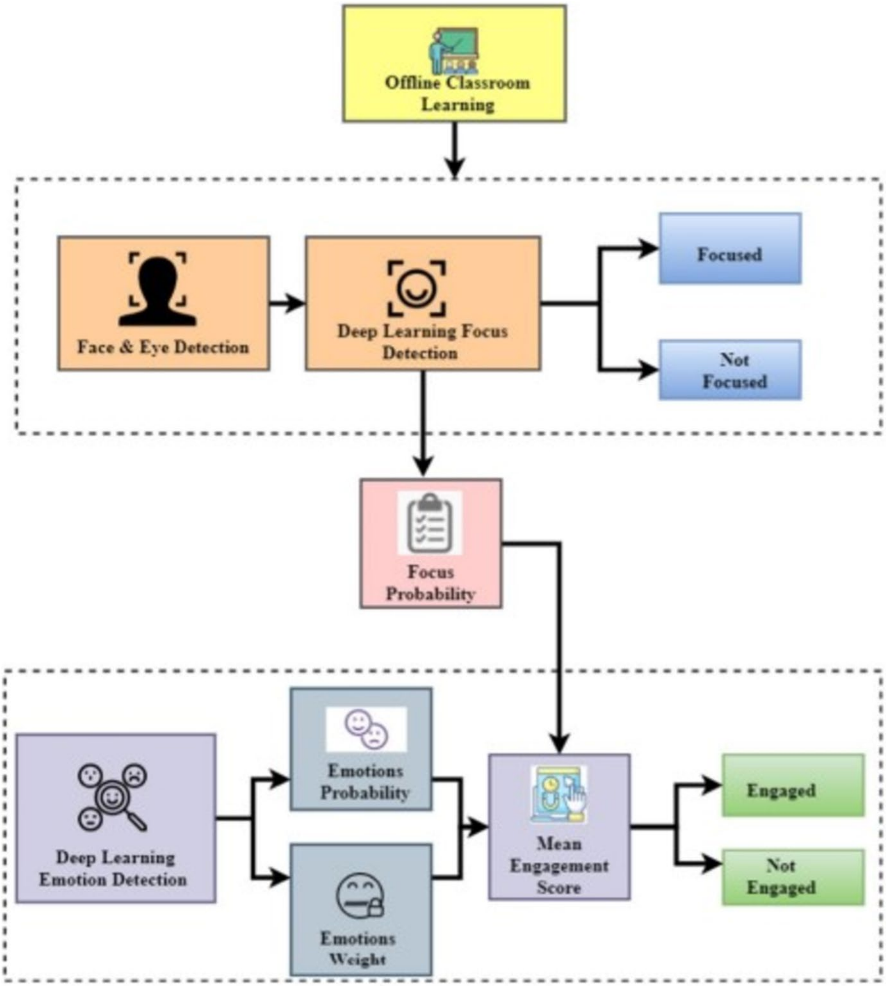


Fig. 8 Student Engagement Detection Model (Akila et al., 2024, p.6872)

4.3.2 A new era in education with artificial neural networks (ANN): Image recognition and learning experiences

Topic 13 (Neural Networks and Image Recognition) has been studied much more in the last decade than other topics (Tables 4 and 5). In fact, it constitutes 13% ($n=1035$) of the studies conducted in this decade. It has increased about 5 times compared to the previous decade.

There have been great developments in the field of Neural Networks and Image Recognition in the last 10 years. The use of these systems in educational technologies may explain the focus on this topic. Different types of neural networks, such as convolutional neural networks (CNNs), are used for image recognition (Galić et al., 2024; Hijazi et al., 2015). Today, image data continues to increase. Currently, video analytics in education is an important research topic (Hu et al., 2023; Khalil et al., 2023; Solanki et al., 2023). Neural Networks and Image Recognition technologies can be used to analyze students' interactions with educational video content. Through these technologies, for example, it can be identified which parts of the video lecture students watch repeatedly and in which parts of the video lecture they lose their attention. In this way, their reactions to the video lecture can be interpreted.

The availability of large amounts of labeled data sets may also have played a role in the researchers' focus on this topic in AIED studies. Analyzing this growing data has an important role in the field of education to improve the learning experience of students/teachers/preservice teachers. For example, to improve the professional skills of prospective or novice teachers, image recognition and neural networks can be used to analyze images and provide feedback systems for them. Student engagement and attention are also very important for learning. The ability to determine students' attention and participation with facial recognition has led to much research on this topic (Hsu et al., 2023; Villegas-Ch et al., 2023a, b). For example, Akila et al. (2024) developed a model based on deep learning-student attention recognition framework (DL-SARF) classifier for the identification of students' engagement (Fig. 8). The model identifies eye regions in faces; "an engagement level is determined by examining just the eye region of a person's face, and a classifier aids in this determination" (Akila et al., 2024, p.6871). With the widespread use of distance education in the Covid-19 period, AI tools can be used for tracking student engagement or working experiences in a virtual laboratory. During virtual lessons, these technologies can be used to analyze students' emotional states during the lesson (expressions such as confusion, difficulty, etc.), give feedback to the teacher by tracking students' facial and eye movements (boredom, participation, etc.), or focus on the material on the screen. In the final analysis, ANN and image recognition tools are an integral part of online and offline education systems (Akila et al., 2024; Prosviryakova et al., 2024).

In the assessment phase, systems that provide automatic assessment (e.g., by recognizing students' handwriting, visual projects, or drawings) can be used (Ahmad et al., 2022). In this way, automatic assessment and feedback can be efficient. Advances in technology (software and hardware systems), the accumulation of large amounts of data, and the changes that Covid-19 has brought about in our lives have been the driving forces behind the trend of this topic.

4.3.3 Innovative approaches in education with deep learning: Personalization, assessment, and beyond

Topic 11 (Deep Learning Approaches in Teaching) has been one of the top three trending topics in two out of six decades (Tables 4 and 5). Deep Learning Approaches, which are the lower layer of Neural Networks, facilitate recognizing objects, understanding texts, and predicting results. The reason why Deep Learning Approaches in Teaching may be one of the most used topics is that deep learning is used and applied to different areas of educational technologies, learning and instruction (e.g., smart classrooms, science teaching, music education, physical education, and language learning) (Hu, 2023; Oraif, 2024; Wang & Shi, 2024; Zhang et al., 2024a, b), and educational administration (Sliwka et al., 2024). On the other hand, the recent popularity of Topic 11 may be related to the fact that some institutions worldwide have started to prefer blended learning with online tools based on deep learning approaches (e.g., ChatGPT using a transformer-based deep learning algorithm) after the COVID-19 pandemic (Wu et al., 2024; Sakib et al., 2024). Specifically, personalized learning environments used in distance education applications, which gained momentum especially during the Covid-19 period, may have made deep learning approaches in teaching a trending topic with applications such as chatbots that provide conversational support as virtual assistants or "virtual teaching assistants" used in courses.

Analyzing large amounts of student data enables the creation of personalized learning environments based on students' characteristics, previous performance, and learning preferences. Instructional content, teaching materials, etc., can be presented according to the characteristics of the students. Adaptive systems based on students' characteristics are increasingly being investigated and widely used in educational practice. Again, deep learning plays an important role in the field of assessment and evaluation, which is an indispensable part of instruction, and can provide feedback to students by evaluating their homework and exams. It also provides educators with information about student progress by continuously monitoring students and allows them to intervene when necessary. In this way, students can be prevented from failing classes, with positive effects on students learning outcomes (Cantú-Ortiz et al., 2020).

4.3.4 The power of machine learning in education: From student performance prediction to customized recommendations

Topic 6 (Machine Learning in Data and Performance Prediction) shows an increase of approximately six times compared to the previous decade. Machine learning is a sub-branch of AI; its algorithms can analyze student behaviors along with their performances and suggest customized paths for each student according to the interest of the students and the subject they have difficulty in. Basic knowledge of machine learning is listed among the AIED competencies (Lameras & Arnab, 2021). In addition, by predicting the success of the students, it can predict and intervene in advance for students who are likely to fail the course or are at risk of dropping out. Topics such as predicting success with machine learning and identifying at-risk students have come up in the literature. Further, the ability to analyze large-scale data with machine learning algorithms may have led this topic to be studied as a trend.

4.4 Assessment and monitoring in education (RQ2, RQ3)

4.4.1 Measuring student achievement in education: The role of AI in assessment and monitoring

Student Performance Assessment is the second topic with the highest average number of publications by year (Fig. 7). Determining student performance is important for educators, administrators, and education policy makers. It can help teachers to evaluate not only students' performance but also the efficiency of curricula and teaching methods, paving the way for changes to be made accordingly. Therefore, an important variable such as student achievement, investigated for years, has become a trending topic in AIED studies. Monitoring student achievement is important in ways such as eliminating deficiencies in learning, increasing the quality of teaching, improving learning, and providing feedback to students. Moreover, thanks to AI technologies, student performance can be monitored and personalized learning environments can be provided, and students' future performance can be determined based on data, which has been the driving force for researchers to focus on this issue. Additionally, AI-based personalized learning platforms are tools for boosting students' academic performance (Xia et al., 2022). In a similar perspective, identifying the amount of student participation in LMSs used in distance education is part of students' performance evaluation. The evaluation of student activities on these technological platforms can guide teachers in terms of student performance assessment.

4.4.2 Neural network algorithms in robotics and detection: Innovations in education pioneered by artificial neural networks

Neural Network Algorithms in Robotics and Detection is among the topics with the highest average number of publications per year (Fig. 7). Robotics applications have frequently been used in education recently to improve students' skills in areas such as problem solving, computational thinking, algorithmic thinking, and programming. Considering the use of robotics sets from kindergarten to university and a large amount of research on robotics, it is highly likely that this topic will attract researchers' attention (Su et al., 2023b).

4.5 Challenges and future directions (RQ4)

We have found in the current study that some trend topics received little interest. We think they are likely to attract researchers' interest more and be trend topics as AI is gets more popular and is applied in different branches of education along with various other sectors. For example, currently, AI in Medical Education is the lowest topic with the lowest average number of publications per year (Fig. 7). This is supported by Zhang et al. (2024a, b). Their systematic review showed that despite the benefits, the application of AI in medical education is facing certain limitations such as technical incompatibilities between the current systems and the developed AI-based tools, poor interactivity, and concerns about the effectiveness of AI in medical education. Moreover, the use of AI technologies in medical education can be costly. The use of AI in medical education requires special expertise in terms of instruction, technology development, and content design. Still, AI in Medical Education is one of the trending topics. Naqvi et al. (2024) predicted that the integration of AI into medical education would have a significant effect on forming the prospects of healthcare. Hence, the positive attitude of scholars towards AI in medical education is likely to attract researchers' interest more in the future.

In a similar vein, the interest in "Adaptive E-Learning and Personalized Education" (2.53%) suggests that future trends may include the development of AI systems that cater to individual learning styles and needs. Research might explore more dynamic and responsive educational technologies that adjust content and instruction in real-time based on student performance and engagement. Very recent studies regard effective integration of adaptive technologies as positive and predict they will help increase the inclusiveness and effectiveness of education (Omar et al., 2024; Santos et al., 2024; Yekollu et al., 2024).

Two final examples can be seen with the sections "Language Processing and Problem Solving" (6.27%) and "Deep Learning Approaches in Teaching" (6.27%). These point towards future research in developing AI tools that aid language learning and cognitive skill development. As AI's capabilities in natural language processing and deep learning continue to advance, these areas are expected to grow, leading to more intelligent tutoring systems and AI-driven educational platforms that support complex problem-solving. Seeing that the current large-scale language models based on deep learning in natural language processing tasks are being questioned in terms of their optimization and new models are on the way (Mei et al., 2024), it can be expected that prospective studies will test their efficiency and report the contributions of new models.

5 Conclusion

AIED encompasses a broad and varied range of academic research areas. The sheer volume of studies within this domain makes it difficult to conduct an exhaustive analysis. Furthermore, there is a noticeable shortage of studies offering an all-encompassing review of the principal trends and themes within AI in education. This study seeks to address this gap by employing text mining methods to explore the development of AI-related educational research over the years. To achieve this, it analyzed 11,027 articles related to education from the Scopus database, spanning from 1967 to 2023. The analysis revealed a marked surge in AI-related educational research starting from 2014, accounting for 73% of all such publications. In the last thirty years, there has been an increasing focus on AI research, especially in the studies carried out with engineering students and published in the form of conference papers. When the distribution by countries is examined, the United States and China stand out as the countries with the highest publication output. As for the subject areas, social sciences and computer science are the main fields that constitute the majority of the publications.

Using the LDA model, 14 different topics were identified for 11,027 articles. Among these, topics such as "AI for Engineering Students", "Neural Networks and Image Recognition", and "Research Review in Educational Study" are particularly highlighted.

To identify topic trends over time, the three topics with the highest number of articles for each decade were determined. Excluding the periods of 2014–2023 and 1967–1973, "AI for Engineering Students" consistently emerged as the top-rated topic. This analysis highlights the significant and ongoing focus on "AI for Engineering Students" within AIED studies over the years, except during the specified periods.

Throughout the decades, key topics have consistently emerged as focal points of research, including "Engineering Students", "Student Performance Assessment", "Neural Network Algorithms in Robotics and Detection", "Language Processing and Problem Solving", and "Neural Networks and Image Recognition". These subjects have not only formed the backbone of relevant works but are also gaining significant momentum. Specifically, "Engineering Students", "Performance Assessment", "Neural Network Algorithms in Robotics and Detection", "Language Processing and Problem Solving", and "Deep Learning Approaches in Teaching" are rapidly advancing fields, attracting increasing attention and development.

In conclusion, this analysis, using text mining, presents a comprehensive picture of the use of AI in educational research and the trends in related subjects over the last 56 years of research within the scope of AIED. The results of the research are expected to guide the prospective research in the field of AIED by addressing trend topics and gaps in the literature. For instance, research can be conducted on topics that show significant increases according to the analysis, including studies on the specific applications of these topics. Additionally, previously unaddressed and unexplored areas related to AI in education can be addressed in future studies.

6 Future directions and suggestions

One significant finding of this study is the rapidly increasing number of studies in the field of AIED. However, most of these studies have been carried out from technological point of view. Thus, studies should be based on a conceptual framework that takes pedagogical approaches into account and even harmonizes itself with them.

AI skills are listed among the technical skills of the twenty-first century, and it is indispensable to know about AI to make full of use it (Ng et al., 2021). Both daily and academic activities require AI literacy. Thus, AI literacy should be part of undergraduate education. We recommend that AI courses be integrated into the undergraduate curricula not only in engineering education but also in various other disciplines so that students can succeed well in their future careers. Therefore, extensive studies on how to incorporate AI into undergraduate curricula are required.

On the other hand, algocracy (governance by algorithms) and datification in the context of the information about students could be considered for the efficiency of the academic and administrative services at university. In brief, research should be expanded to unlock the potentials of big data in education.

7 Limitations

The study broadly covers AIED in various educational contexts without delving into the specific nuances of different educational levels, such as primary, secondary, and higher education. Subsequent research endeavors could investigate how the efficacy of AI applications differs across these diverse environments.

Furthermore, it is important to note that our study is constrained by the information extracted solely from the Scopus database spanning the years 1967 to 2023. Future studies could enhance comprehensiveness by extending the scope of data sources, considering additional features such as bi-grams or co-occurrence patterns, and incorporating more recent information.

Abbreviations *ACO*: Ant colony optimization; *AI*: Artificial intelligence; *AIED*: Artificial intelligence in education; *GA*: Genetic algorithms; *IT*: Information technologies; *ITS*: Intelligent tutoring systems; *ML*: Machine learning; *NLP*: Natural language processing; *PSO*: Particle swarm optimization; *SPM*: Sequential pattern mining; *SPOC*: Small private online course

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest None.

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